



[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

- There are 3 (Three) questions. Answer all 03 questions. All questions are of values indicated on the right-hand margin.

Q1.

- a) Suppose Alice and Bob are communicating over a TCP/IP network, with a **router (T)** and a **link-layer switch (S)** positioned between the two hosts. Each of these devices (hosts, router and switch) implements a **different set of TCP/IP layers**, reflecting their distinct functionalities.
- With the aid of **diagrams**, demonstrate the **encapsulation and de-encapsulation** processes. [3]
 - Among the following protocols **IP, UDP, Ethernet, SMTP, TCP, and HTTP**, which protocols are implemented in the **intermediate nodes** and **why**. [1]
 - When Alice and Bob communicate **via email**, to **identify** the receiving process, **how many pieces of information** need to be specified and what are they? Illustrate with an **example**. [2]
- b) Suppose, **Host A** is sending a **message** of length **500 KB** to **Host B**. Bandwidths of two segments are **$R_1 = 5 \text{ Mbps}$** and **$R_2 = 1 \text{ Mbps}$** . Distances of two segments are **$D_1 = 2 \text{ km}$** and **$D_2 = 1 \text{ km}$** respectively. If **$2 \times 10^8 \text{ m/s}$** is the **propagation speed** in medium, find out the **end-to-end delay** from Host A to Host B while **processing and queuing delays** inside the router is **5 msec** and **10 msec** respectively. [3]
- ```
graph LR; HA[Host A] ---|D1, R1| R[Router, R]; R ---|D2, R2| HB[Host B]
```
- c) Suppose that Alice and Bob share a **1 Mbps link**, where every user in that link reserves **10 Kbps**, then in **circuit switching** technology **how many users** may be served if each user is **active 10%** of the time? **Justify** your answer. [2]
- d) Suppose that  **$R_s$  bits/sec** is the **link capacity** between Alice and S, whereas  **$R_T$  bits/sec** is the capacity between Bob and T. If the value of  **$R_T$**  is **2 Mbps** and the value of  **$R_s$**  is **5 Mbps**. What is the **throughput**? [1]

### Q2.

- a) **HTTP** and **SMTP** are both file transfer protocols and have many common characters; for example, they both are client-server protocols and run on top of TCP. List **any two differences** between HTTP and SMTP? [2]



- b) Suppose that a *Web server* and an *FTP server* run on **port number 80** and **21** respectively in *Host C*, which is currently receiving requests from **two different Hosts, A and B**. Browsers of **Host A** run on **port number 5555** and **Host B** is requesting FTP service using **port number 6666**. With the help of a **diagram** explain **how** does Host C directs segments to the **appropriate socket**? What **tuples (IP: Port)** are being used here? [3]
- c) I. Explain how *Symmetric key cryptography* and *Public Key cryptography* work and what are their advantages & disadvantages. [2]  
II. What are the differences between Message *Confidentiality* and message *Integrity*? [1]  
III. Can you have confidentiality without integrity or can you have integrity without confidentiality? Justify your answer. [1]

### Q3.

- a) Consider a client accessing a **base HTML page** with **four (4) URL references** to images to be displayed in the page. Assume that it takes **time T1** to establish a TCP connection, **time T2** to retrieve the base page, and **time T3** to retrieve each of the images. **Ignore** time to close a TCP connection, and time to send HTTP request packets. [3]  
I. **Calculate** the **total time (in terms of T1, T2 and T3)** it takes to retrieve the base page and its embedded images for **non-persistent and persistent HTTP**.  
II. **Draw** the corresponding **sequence diagram (steps)** for the above scenario.
- b) Suppose **Rahman**, with a Web-based e-mail account (rahman@gmail.com), sends an email to **Ali** (ali@hotmail.com), who accesses his email using **IMAP**. With a **diagram** explain how the email moves from **Rahman's host** to **Ali's mailbox**. Be sure to **identify the application-layer protocols** that are used to move the email between the two hosts. [3]
- c) Consider a scenario with **Host A**, **local DNS server B**, and the **rest of the DNS infrastructure**. Host A asks Host B to resolve **www.eLMS.uiu.ac.bd**. **List the steps** taken to resolve **www.eLMS.uiu.ac.bd** and respond to A. In each step **identify who B contacts**. [3]

End of Paper – Thank You